

An explicit second-order bound for Sidon sets

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Abstract

Let $F(N)$ be the largest size of a Sidon subset of $\{1, \dots, N\}$, with diagonal pair sums included. We prove $F(N) \leq \sqrt{N} + (2\sqrt{2}/3)N^{1/4} + 1$ for every integer $N \geq 120^4$. The proof uses weighted Erdős–Turán energy with $h(t) = 2(1-t)\mathbf{1}_{[0,1]}(t)$ and an exact boundary correction of $2/3$; an elementary contraction estimate makes the remainder explicit. Reported coefficients run from Lindström’s 1 through Hou–Zhao’s $0.94349\dots$ to unrefereed repository claims near 0.94324 and 0.94301 ; the present coefficient is $0.942809\dots$ Erdős’s conjecture $F(N) = \sqrt{N} + O_\varepsilon(N^\varepsilon)$ for every $\varepsilon > 0$ remains untouched.

Keywords: Sidon sets; distinct differences; finite energy; explicit bounds; boundary correction.

1 Introduction

A Sidon set of integers has distinct unordered two-term sums, including sums in which the two summands agree. Write $F(N)$ for the largest cardinality of such a set in $\{1, \dots, N\}$. Translation gives the same maximum on $\{0, \dots, N-1\}$. The problem of bounding $F(N)$ goes back to Erdős and Turán [ET41], who proved $F(N) \leq \sqrt{N} + O(N^{1/4})$. Lindström [Lind69] gave the explicit bound $F(N) < \sqrt{N} + N^{1/4} + 1$, valid for every positive integer N . Cilleruelo [Cil10] reduced its additive constant to $1/2$. The reproductions and proofs in Balogh–Füredi–Roy [BFR23] also record these two explicit statements.

Table 1 gives a source-qualified progression of second-order coefficients at the scale $\sqrt{N} + cN^{1/4}$. Where a source states a uniform $O(1)$ remainder or a stronger form, the table indicates it; no additional remainder assertion is assigned here to an announcement. A coefficient with an unspecified $O(1)$ is not a printed numerical pair of additive constant and onset. Publication years and preprint years are listed separately when they differ. The table includes unrefereed announcements and repository claims as such, and does not certify their computational certificates.

Carter–Hunter–O’Bryant [CHO25] state a diameter bound $k^2 - 1.96365k^{3/2} - O(k)$ for a k -element Sidon set, yielding the safe cardinality coefficient 0.98183 in the table. The 2025 collaboration of Carter, Georgiev, Gómez-Serrano, Hunter, O’Bryant, Tao and Wagner reported a diameter coefficient 1.9526463099204112 ; half that value explains the rounded safe entry 0.97633 [AE25]. These comparisons concern asymptotic coefficients: the difference between diameter $N-1$ and interval parameter N does not affect that coefficient, but must be retained when asserting a specified additive constant.

The present note gives a complete proof with the numerical additive constant and onset printed in the theorem. It uses the autocorrelation of a decreasing linear weight. An exact correction on a half-line has excess mass $1/3$; joining two copies gives the boundary term $2/3$. A quantitative elementary contraction controls the tails. The auxiliary measure on a finite interval is signed, which is permitted because the energy comparison is an L^2 Cauchy–Schwarz inequality. No theorem about renewal processes is needed.

Erdős’s conjecture asks for $F(N) = \sqrt{N} + O_\varepsilon(N^\varepsilon)$ for every $\varepsilon > 0$; see the problem record [EP30]. The estimate proved here retains the exponent $1/4$ and does not resolve that conjecture. The source comparison above makes no claim that the displayed list is exhaustive or that the method has a universal optimality barrier.

2 Statement

A finite set of integers A is a **Sidon set** if the map

$$\{(a, b) \in A^2 : a \leq b\} \longrightarrow \mathbb{Z}, \quad (a, b) \longmapsto a + b$$

Source	Coefficient c	Form and status
Erdős–Turán (1941) [ET41]	unspecified	$O(N^{1/4})$; journal article
Lindström (1969) [Lind69]	1	additive +1, every $N \geq 1$; journal article
Cilleruelo (2010) [Cil10]	1	additive +1/2, every $N \geq 1$; journal article
Balogh–Füredi–Roy (2021/2023) [BFR23]	0.998	no additive term for all sufficiently large N ; journal article
O’Bryant (2022/2024) [OB24]	0.99703	no additive term for all sufficiently large N ; journal article
Carter–Hunter–O’Bryant (2023/2025) [CHO25]	0.98183	$O(1)$ remainder; journal article
AlphaEvolve collaboration (2025) [AE25]	0.97633	rounded safe coefficient; unrefereed announcement
Hou–Zhao v1 (July 2026) [HZ26]	0.9460052...	$O(1)$ remainder; preprint
Hou–Zhao v3 (September 2026) [HZ26]	0.9434925907...	$O(1)$ remainder; preprint
JoshMadeiros (August 2026) [Josh26]	≈ 0.94324445	$c < 0.94325$; $O(1)$; unrefereed forum claim
Wu, q1 (August 2026) [Wu26]	≈ 0.94324253	$c < 0.94325$; $O(1)$; unrefereed repository claim
Wu, q2 (August 2026) [Wu26]	≈ 0.94300617	$c < 0.94301$; $O(1)$; unrefereed repository claim
This note	$2\sqrt{2}/3$	additive +1 for every $N \geq 120^4$; review status below

Table 1: Selected coefficient history. The initial date is the preprint or announcement date when a later journal date is also displayed. Decimal entries retain the scope and status of their cited sources.

is injective. In particular, the pairs (a, a) are included: equality $a + b = c + d$, with $a \leq b$ and $c \leq d$, requires $a = c$ and $b = d$.

Theorem 2.1. *For every integer $N \geq 207\,360\,000$ and every Sidon set $A \subseteq \{1, \dots, N\}$,*

$$|A| \leq \sqrt{N} + \frac{2\sqrt{2}}{3}N^{1/4} + 1. \quad (2.1)$$

All powers and square roots in this document are positive real ones. In particular, there is no requirement that N be a fourth power. The onset is $207\,360\,000 = 120^4$.

Lemma 2.2 (equivalent difference convention). *The Sidon condition is equivalent to*

$$a - b = c - d \neq 0, \quad a, b, c, d \in A \implies (a, b) = (c, d). \quad (2.2)$$

Here the differences are indexed by ordered pairs, and zero differences are excluded. Equivalently, each positive integer is the difference $a - b$, with $a, b \in A$ and $a > b$, at most once.

Proof. Suppose first that pair sums, including diagonal sums, are unique. The equality $a - b = c - d \neq 0$ gives $a + d = c + b$. Equality of the two unordered pairs with multiplicities gives either $a = c, d = b$, as required, or $a = b, d = c$. The latter possibility contradicts the nonzero difference.

Conversely, suppose (2.2) holds, and let $a + b = c + d$. If $a = c$, then $b = d$. Otherwise $a - c = d - b \neq 0$, so (2.2), applied to the ordered pairs (a, c) and (d, b) , gives $a = d, c = b$. Thus the unordered pairs with multiplicities agree. Restricting both pairs to increasing order proves the stated Sidon condition, including when a pair is diagonal. Finally, a negative difference becomes a positive one on reversing both ordered pairs; hence uniqueness of positive differences is equivalent to (2.2). $\square$

Translation preserves these conditions. We may consequently replace A by $A - 1 \subseteq \{0, \dots, N - 1\}$ when proving (2.1).

3 Kernel and energy

Lebesgue measure on the real line is denoted by dt . For an integrable function v and a measure μ , write

$$(v * \mu)(x) = \int v(x-t) d\mu(t)$$

whenever the integral exists. A finite signed real measure means one with finite total variation $\|\mu\|_{\text{TV}} = |\mu|(\mathbb{R})$.

Set

$$h(t) = 2(1-t)\mathbf{1}_{[0,1]}(t), \quad \tilde{h}(t) = h(-t), \quad f = h * \tilde{h}. \quad (3.1)$$

The specified value $h(0) = 2$ will be convenient in a pointwise identity.

Lemma 3.1 (kernel and Cauchy inequality). *The function f is even and*

$$f(x) = \begin{cases} \frac{4}{3} - 2|x| + \frac{2}{3}|x|^3, & |x| \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (3.2)$$

It is nonnegative, nonincreasing on $[0, \infty)$, and satisfies

$$f(0) = \|h\|_2^2 = \frac{4}{3}, \quad \int_{\mathbb{R}} h(t) dt = 1, \quad \int_{\mathbb{R}} f(t) dt = 1. \quad (3.3)$$

For finite signed real measures define

$$E(\mu, \nu) = \iint f(x-y) d\mu(x) d\nu(y).$$

Then

$$E(\mu, \nu) = \int_{\mathbb{R}} (h * \mu)(t)(h * \nu)(t) dt, \quad E(\mu, \mu) \geq 0, \quad (3.4)$$

and

$$|E(\mu, \nu)|^2 \leq E(\mu, \mu)E(\nu, \nu). \quad (3.5)$$

Proof. The autocorrelation is even by a change of variable. For $0 \leq x \leq 1$, its value is

$$4 \int_0^{1-x} (1-s-x)(1-s) ds = \frac{4}{3} - 2x + \frac{2}{3}x^3;$$

for $x > 1$ the two supports have no overlap of positive length. This proves (3.2). Nonnegativity also follows directly from the integral of the two nonnegative factors. On $[0, 1]$ the derivative is $-2 + 2x^2 \leq 0$, and outside this interval the function is zero. Integrating h and h^2 gives (3.3); the mass of f is $(\int h)^2 = 1$ by Tonelli's theorem.

For any finite signed measure, Cauchy–Schwarz with respect to $|\mu|$ gives

$$\int |h * \mu|^2 \leq \|\mu\|_{\text{TV}} \int \int |h(t-x)|^2 d|\mu|(x) dt = \|\mu\|_{\text{TV}}^2 \|h\|_2^2.$$

Thus $h * \mu \in L^2$. Moreover, $\int |h(t-x)h(t-y)| dt \leq \|h\|_2^2$, so the corresponding triple integral against $|\mu|$ and $|\nu|$ is finite. Fubini's theorem is therefore applicable. Its inner integral is $f(x-y)$, which proves (3.4). Applying the usual Cauchy–Schwarz inequality in real L^2 proves (3.5). This argument allows either measure to be signed. $\square$

4 The half-line correction

Let U be the probability measure with density $v_1 = \mathbf{1}_{[0,1]}$, and let $U^{*0} = \delta_0$. Define

$$g_+ = \frac{1}{2} \sum_{n=0}^{\infty} U^{*n}. \quad (4.1)$$

The following construction proves everything needed about this measure; no renewal theorem is used.

Lemma 4.1 (local finiteness and exact potential). *The measure g_+ is locally finite, supported on $[0, \infty)$, and*

$$h * g_+ = \mathbf{1}_{[0, \infty)}, \quad (f * g_+)(x) = 1 \quad (x \geq 0). \quad (4.2)$$

Proof. For $n \geq 1$, let v_n be the density of U^{*n} . Convolution gives representatives satisfying, for $t \geq 0$,

$$0 \leq v_n(t) \leq \frac{t^{n-1}}{(n-1)!}. \quad (4.3)$$

For $n = 1$ this is the bound $v_1 \leq 1$. If it holds at n , then

$$v_{n+1}(t) = \int_0^{\min(1,t)} v_n(t-s) ds \leq \int_0^t \frac{(t-s)^{n-1}}{(n-1)!} ds = \frac{t^n}{n!}.$$

For every fixed $R \geq 0$, it follows that $U^{*n}([0, R]) \leq R^n/n!$. The sum in (4.1) consequently has finite mass on every bounded interval.

For a direct proof of the first identity in (4.2), put $H = \mathbf{1}_{[0, \infty)}$. The definitions give the pointwise identity $h/2 = H - U * H$. Thus each finite partial sum telescopes:

$$\frac{1}{2} \sum_{n=0}^m h * U^{*n} = H - U^{*(m+1)} * H. \quad (4.4)$$

For $t \geq 0$, the last term on the right is between zero and $t^{m+1}/(m+1)!$, by (4.3), and this tends to zero. For $t < 0$, both sides vanish. The nonnegative partial sums on the left converge to $h * g_+$, proving the identity. This also fixes its value at $t = 0$.

Tonelli's theorem allows association of the nonnegative convolutions. For $x \geq 0$,

$$(f * g_+)(x) = (\tilde{h} * (h * g_+))(x) = \int_{-1}^0 \tilde{h}(y) H(x-y) dy = \int_{-1}^0 \tilde{h}(y) dy = 1.$$

All the integrals against g_+ here are locally finite because their kernels have compact support. □

Write

$$g_+ = \frac{1}{2} \delta_0 + \frac{1}{2} u(t) \mathbf{1}_{[0, \infty)}(t) dt, \quad u(t) = \sum_{n=1}^{\infty} v_n(t) \quad (t \geq 0). \quad (4.5)$$

Use the right-continuous representative of u . In particular,

$$u(0) = 1, \quad u(t) = e^t \quad (0 \leq t < 1), \quad u(1) = e - 1. \quad (4.6)$$

The left limit at 1 is e , and must not be substituted for $u(1)$ in the recurrence below. To justify these regularity claims, the series in (4.5) converges uniformly on compact intervals by (4.3). For $n \geq 2$, the density $v_n(t) = \int_{t-1}^t v_{n-1}(s) ds$, with densities set to zero for $s < 0$, is continuous. Only v_1 has a jump at 1. For $0 \leq t < 1$, the induction proving (4.3) gives equality. At $t = 1$, $v_1(1) = 0$ and $v_n(1) = 1/(n-1)!$ for $n \geq 2$. These observations prove (4.6) and the stated representative convention.

Lemma 4.2 (quantitative convergence of the density). *For almost every $t \geq 0$,*

$$|u(t) - 2| \leq (e-1)(3/4)^{\lfloor t \rfloor} < 2(3/4)^{\lfloor t \rfloor}. \quad (4.7)$$

Proof. Summing the convolution recurrence for the densities yields $u = v_1 + U * u$. Thus

$$u(t) = \int_{t-1}^t u(s) ds \quad (t \geq 1). \quad (4.8)$$

This includes $t = 1$ with the right-continuous value in (4.6). On $(1, \infty)$, this identity makes u locally absolutely continuous, and $u'(t) = u(t) - u(t-1)$ almost everywhere. Fix an integer $n \geq 1$. Solving this first-order equation on $[n, n+1]$ gives

$$u(n+x) = e^x u(n) - \int_0^x e^{x-y} u(n-1+y) dy \quad (0 \leq x \leq 1).$$

Since $u(n) = \int_0^1 u(n-1+y) dy$, this becomes

$$u(n+x) = \int_0^1 K_x(y) u(n-1+y) dy, \quad K_x(y) = e^x - \mathbf{1}_{\{y \leq x\}} e^{x-y}. \quad (4.9)$$

For $0 \leq x, y \leq 1$, the kernel is nonnegative, and

$$\int_0^1 K_x(y) dy = e^x - \int_0^x e^{x-y} dy = 1.$$

For $y \in [1/2, 1]$, it also satisfies $K_x(y) \geq 1/2$. Indeed, if $y > x$, its value is $e^x \geq 1$. If $y \leq x$, its value is $e^{x-y}(e^y - 1) \geq e^{1/2} - 1 > 1/2$, using the exponential series.

Let $[m_n, M_n]$ be the smallest closed interval containing the essential range of u on $[n, n+1]$. Here essential extrema ignore sets of Lebesgue measure zero. Initially $[m_0, M_0] = [1, e]$. Equation (4.9) first implies that these closed intervals are nested. For the width estimate, every $K_x(y) dy$ contains the common measure

$$w(y) dy = \frac{1}{2} \mathbf{1}_{[1/2, 1]}(y) dy$$

of mass $1/4$. Put $c_n = \int_0^1 w(y) u(n-1+y) dy$. The remaining nonnegative measure has mass $3/4$, so every value in (4.9) lies between $c_n + (3/4)m_{n-1}$ and $c_n + (3/4)M_{n-1}$. Consequently,

$$[m_n, M_n] \subseteq [m_{n-1}, M_{n-1}], \quad M_n - m_n \leq \frac{3}{4}(M_{n-1} - m_{n-1}).$$

The nested closed intervals therefore meet in a single real number c , and

$$|u(t) - c| \leq (e-1)(3/4)^{\lfloor t \rfloor} \quad (4.10)$$

almost everywhere. This also provides a uniform essential bound and essential uniform convergence to c as $t \rightarrow \infty$.

For $t > 1$, the atom in (4.5) contributes nothing to $h * g_+(t)$, so Lemma 4.1 gives

$$1 = \frac{1}{2} \int_0^1 h(s) u(t-s) ds.$$

Letting $t \rightarrow \infty$ in this integral, (4.10) and $\int h = 1$ imply $1 = c/2$. Thus $c = 2$. Finally, $e < 3$: in its series $e = 2 + \sum_{j \geq 2} 1/j!$, the tail is strictly smaller than $\sum_{j \geq 2} 2^{-(j-1)} = 1$. This proves (4.7). $\square$

Lemma 4.3 (size, tail, and mass of the correction). *Define the signed measure supported on $[0, \infty)$*

$$q = g_+ - \mathbf{1}_{[0, \infty)}(t) dt = \frac{1}{2} \delta_0 + r(t) \mathbf{1}_{[0, \infty)}(t) dt, \quad r(t) = \frac{1}{2} u(t) - 1. \quad (4.11)$$

It is a finite signed measure, and, with $\alpha = \log(4/3)$,

$$|r(t)| \leq (3/4)^{\lfloor t \rfloor} \quad \text{almost everywhere}, \quad \|q\|_{\text{TV}} \leq \frac{9}{2}, \quad (4.12)$$

$$|q|((L, \infty)) \leq 6e^{-\alpha L} \quad (L \geq 1), \quad (4.13)$$

and

$$q(\mathbb{R}) = \frac{1}{3}. \quad (4.14)$$

Proof. The first bound follows from Lemma 4.2. Integration on unit intervals then gives

$$\|q\|_{\text{TV}} \leq \frac{1}{2} + \sum_{n=0}^{\infty} (3/4)^n = \frac{9}{2}.$$

For $L \geq 1$, the atom at zero is outside (L, ∞) , and

$$|q|((L, \infty)) \leq \sum_{n=\lfloor L \rfloor}^{\infty} (3/4)^n = 4(3/4)^{\lfloor L \rfloor} \leq \frac{16}{3} e^{-\alpha L} \leq 6e^{-\alpha L}.$$

It remains to compute the signed mass, not just its absolute bound. For $s > 0$, Tonelli's theorem and the product rule for Laplace transforms of nonnegative convolutions give

$$\begin{aligned} \int e^{-st} dg_+(t) &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\int_0^1 e^{-st} dt \right)^n \\ &= \frac{1}{2(1 - (1 - e^{-s})/s)}. \end{aligned} \quad (4.15)$$

The geometric series converges since $0 < \int_0^1 e^{-st} dt < 1$. Taylor's formula for the exponential, as $s \downarrow 0$, gives

$$\phi(s) := 1 - \frac{1 - e^{-s}}{s} = \frac{s}{2} - \frac{s^2}{6} + O(s^3).$$

This use of $O(s^3)$ concerns only a limit: for $0 < s \leq 1$, Taylor's remainder after the cubic term of e^{-s} has absolute value at most $s^4/24$, which supplies the displayed bound directly. Subtracting the Laplace transform $1/s$ of half-line Lebesgue measure gives

$$\int e^{-st} dq(t) = \frac{1}{2\phi(s)} - \frac{1}{s} = \frac{s - 2\phi(s)}{2s\phi(s)} \longrightarrow \frac{1}{3}.$$

Because q has finite total variation and is supported on the nonnegative half-line, $|e^{-st}| \leq 1$ there. Dominated convergence for the finite measure $|q|$ shows that the limit is $q(\mathbb{R})$, proving (4.14). $\square$

4.1 Joining two half-lines

Lemma 4.4 (finite-interval energy bound). *Let $L \geq 1$, and let μ be any finite positive measure supported on the closed interval $[0, L]$. If $k = \mu(\mathbb{R})$, then*

$$E(\mu, \mu) \geq \frac{k^2}{L + \frac{2}{3} + 200e^{-\alpha L}}, \quad \alpha = \log(4/3). \quad (4.16)$$

Proof. Let q_L be the pushforward of q by the reflection $t \mapsto L - t$, and set

$$\nu_L = \mathbf{1}_{[0, L]}(t) dt + q + q_L. \quad (4.17)$$

This is a finite signed real measure, with

$$\nu_L(\mathbb{R}) = L + \frac{2}{3}. \quad (4.18)$$

Let g_-^L denote the same reflection of g_+ . As locally finite measures,

$$\nu_L = g_+ + g_-^L - dt. \quad (4.19)$$

Indeed, the two half-line Lebesgue measures sum to full-line Lebesgue measure plus Lebesgue measure on $[0, L]$; endpoints do not change those Lebesgue measures. The atoms of q at 0 and of q_L at L remain included

in (4.17). Since f is compactly supported, (4.19) may be convolved locally with f . Lemma 4.1 and evenness of f give, for $0 \leq x \leq L$,

$$V_L(x) := (f * \nu_L)(x) = 1 + 1 - \int f = 1. \quad (4.20)$$

This holds also at both endpoints. For every real x , nonnegativity and mass one of f , together with Lemma 4.3, imply

$$|V_L(x)| \leq 1 + \|f\|_\infty (\|q\|_{\text{TV}} + \|q_L\|_{\text{TV}}) \leq 1 + \frac{4}{3} \cdot 9 = 13. \quad (4.21)$$

Outside $[0, L]$, the only measure in (4.17) on (L, ∞) is the tail of q ; on $(-\infty, 0)$ it is the reflection of the same tail. In particular there are no excluded endpoint atoms. Thus (4.13) gives

$$|\nu_L|(\mathbb{R} \setminus [0, L]) \leq 12e^{-\alpha L}.$$

By (4.18), (4.20), and (4.21),

$$\begin{aligned} \left| E(\nu_L, \nu_L) - \left(L + \frac{2}{3} \right) \right| &= \left| \int_{\mathbb{R} \setminus [0, L]} (V_L(x) - 1) d\nu_L(x) \right| \\ &\leq 14 \cdot 12e^{-\alpha L} = 168e^{-\alpha L} \leq 200e^{-\alpha L}. \end{aligned} \quad (4.22)$$

Also $E(\mu, \nu_L) = \int V_L d\mu = k$. Lemma 3.1 now gives

$$k^2 \leq E(\mu, \mu) E(\nu_L, \nu_L) \leq E(\mu, \mu) \left(L + \frac{2}{3} + 200e^{-\alpha L} \right).$$

The denominator is positive and $E(\mu, \mu) \geq 0$, proving (4.16). There is no assumption that ν_L is positive. $\square$

5 The discrete inequality

Lemma 5.1 (Sidon energy upper bound). *For every integer $N \geq 1$, every Sidon set $A \subseteq \{0, \dots, N-1\}$, and every real $0 < T \leq N$, writing $k = |A|$ gives*

$$k^2 \leq \left(\frac{N}{T} + \frac{2}{3} + 200e^{-\alpha N/T} \right) \left(T + \frac{4}{3}k \right). \quad (5.1)$$

Proof. Put $L = N/T \geq 1$ and $\mu = \sum_{a \in A} \delta_{a/T}$. Its mass is k , and its support lies in $[0, L]$. In fact its right endpoint is at most $(N-1)/T$; using the larger interval of length N/T is permitted in Lemma 4.4.

Let $D_+(A)$ denote the set of positive differences of elements of A . Lemma 2.2 says that each such difference occurs for exactly one ordered pair. The zero differences are precisely the k diagonal ordered pairs. Evenness of f therefore gives the exact identity

$$E(\mu, \mu) = \frac{4}{3}k + 2 \sum_{d \in D_+(A)} f(d/T) \leq \frac{4}{3}k + 2 \sum_{d=1}^{\infty} f(d/T). \quad (5.2)$$

The inequality uses nonnegativity of f . For every integer $d \geq 1$, monotonicity on the positive half-line implies

$$f(d/T) \leq T \int_{(d-1)/T}^{d/T} f(s) ds.$$

Summing, and using evenness and $\int f = 1$, yields

$$\sum_{d=1}^{\infty} f(d/T) \leq T \int_0^{\infty} f(s) ds = \frac{T}{2}.$$

Only finitely many summands can be nonzero. It follows that $E(\mu, \mu) \leq T + 4k/3$. Combining this with (4.16) proves (5.1). This proof applies also to the empty set and to singleton sets. $\square$

6 Explicit constants and onset

Proof of Theorem 2.1. Let

$$x = N^{1/4}, \quad T = \sqrt{2}x^3, \quad \gamma = \frac{2\sqrt{2}}{3}, \quad \varepsilon = 200e^{-\alpha x/\sqrt{2}}. \quad (6.1)$$

For $x \geq 120$ the interval parameter $L = N/T = x/\sqrt{2}$ is at least one, so Lemma 5.1 applies. Expanding its right side gives $P_\varepsilon(k) \leq 0$, where

$$P_\varepsilon(z) = z^2 - \left(\gamma x + \frac{8}{9} + \frac{4}{3}\varepsilon \right) z - x^4 - \gamma x^3 - \sqrt{2}\varepsilon x^3. \quad (6.2)$$

For clarity, the coefficient of z comes from $(4/3)(x/\sqrt{2} + 2/3 + \varepsilon)$, and the constant term comes from $-(x/\sqrt{2} + 2/3 + \varepsilon)\sqrt{2}x^3$.

We next establish a uniform bound for the error with all numerical comparisons explicit. The integral formula for the logarithm gives

$$\alpha = \int_1^{4/3} \frac{dt}{t} \geq \frac{1}{4}.$$

Since $\sqrt{2} \leq 3/2$, it follows that $\alpha/\sqrt{2} \geq 1/6$. The derivative of $xe^{-x/6}$ is $e^{-x/6}(1-x/6)$, so it is nonpositive for $x \geq 6$. Consequently, for every real $x \geq 120$,

$$\varepsilon x \leq 200xe^{-x/6} \leq 24000e^{-20} < \frac{24000}{2^{20}} = \frac{375}{16384} < \frac{1}{32}. \quad (6.3)$$

Here $e > 2$ follows from its series, and the last comparison is the integer inequality $375 \cdot 32 = 12000 < 16384$. Equivalently, $24000 \cdot 32 = 768000 < 1048576 = 2^{20}$.

This is the reason for the convenient choice $x_0 = 120$, hence $N_0 = 120^4 = 207360000$: it makes the exponent $x_0/6 = 20$ and proves (6.3) by integer arithmetic. This onset is not claimed minimal. The argument is uniform beyond the onset, rather than a check at one point. More explicitly, the controlling envelope

$$H(N) = 200N^{1/4} \exp(-N^{1/4}/6)$$

has derivative

$$H'(N) = 50N^{-3/4} \exp(-N^{1/4}/6) \left(1 - \frac{N^{1/4}}{6} \right) \leq 0 \quad (N \geq 6^4). \quad (6.4)$$

The actual quantity εx is also decreasing there, since $\alpha/\sqrt{2} \geq 1/6$. Thus no larger N can violate the error bound used in (6.3).

Put $y = x^2 + \gamma x + 1$. Using $\gamma^2 = 8/9$, direct expansion gives

$$P_0(y) = \frac{10}{9}x^2 + \frac{\gamma}{9}x + \frac{1}{9} \geq x^2. \quad (6.5)$$

Also $\gamma < 1$, since $\gamma^2 = 8/9 < 1$. For $x \geq 1$, $y \leq x^2 + x + 1 \leq 3x^2$. Hence (6.3) implies

$$\begin{aligned} \varepsilon \left(\frac{4}{3}y + \sqrt{2}x^3 \right) &< \frac{1}{32x} \left(4x^2 + \frac{3}{2}x^3 \right) \\ &= \frac{x}{8} + \frac{3x^2}{64} \leq \frac{11x^2}{64}. \end{aligned} \quad (6.6)$$

The last step uses $x \leq x^2$, valid for $x \geq 1$. Combining (6.2), (6.5), and (6.6) now proves, for every real $x \geq 120$,

$$P_\varepsilon(y) = P_0(y) - \varepsilon \left(\frac{4}{3}y + \sqrt{2}x^3 \right) > \frac{53}{64}x^2 > 0. \quad (6.7)$$

The quadratic P_ε has leading coefficient 1 and strictly negative constant term. Its two real roots have opposite signs: this also follows directly from its positive discriminant and negative root product. Therefore,

on $[0, \infty)$, the condition $P_\varepsilon(k) \leq 0$ places k at or below the unique positive root. Since $y > 0$ and $P_\varepsilon(y) > 0$, that root is strictly smaller than y . We have proved

$$k < x^2 + \gamma x + 1 = \sqrt{N} + \frac{2\sqrt{2}}{3}N^{1/4} + 1.$$

Translating the original set as in Section 2 proves (2.1). In particular, the stated non-strict bound follows from a strict bound on the real right side. $\square$

7 Where $2\sqrt{2}/3$ comes from

Two exact constants determine this coefficient in the retained scalar energy inequality:

$$a = f(0) = \frac{4}{3}, \quad b = 2q(\mathbb{R}) = \frac{2}{3}. \quad (7.1)$$

The first is the diagonal cost per point in (5.2). The second is the sum of the two half-line excess masses in (4.18). Apart from the explicitly controlled exponential error, (5.1) has right side

$$\left(\frac{N}{T} + b\right)(T + ak) = N + bT + \frac{aNk}{T} + abk. \quad (7.2)$$

To read its second-order scale, put $N = x^4$, $T = tx^3$ for a fixed $t > 0$, and $k = x^2 + cx + O(1)$. The terms of order x^3 in the two sides of the inequality are respectively $2cx^3$ and $(bt + a/t)x^3$. The resulting upper coefficient is

$$c(t) = \frac{1}{2} \left(bt + \frac{a}{t}\right) = \frac{t}{3} + \frac{2}{3t}. \quad (7.3)$$

For $t > 0$,

$$bt + \frac{a}{t} - 2\sqrt{ab} = \left(\sqrt{bt} - \sqrt{a/t}\right)^2 \geq 0.$$

Thus the minimum in (7.3) is $\sqrt{ab} = 2\sqrt{2}/3$, attained precisely at $t = \sqrt{a/b} = \sqrt{2}$. This gives the scale chosen in (6.1). For fixed t , the exponential term has argument $-\alpha x/t$, so it contributes no term at this order; Section 6 supplies its fully effective treatment for the selected t .

A precise limitation statement is also possible. It concerns **only the scalar inequalities (5.1), with this kernel and these retained energy estimates**. It does not concern all arguments using this kernel, all boundary corrections, or all Sidon methods.

To prove this limited statement, for each $N > 0$ let $\kappa_N > 0$ be the solution of

$$\kappa_N = \sqrt{N} + \sqrt{ab\kappa_N}. \quad (7.4)$$

Writing $x = N^{1/4}$ and $\gamma = \sqrt{ab}$ gives explicitly

$$\kappa_N = \left(\frac{\gamma + \sqrt{\gamma^2 + 4x^2}}{2}\right)^2 = x^2 + \gamma\sqrt{x^2 + \gamma^2/4} + \frac{\gamma^2}{2}. \quad (7.5)$$

For every $T > 0$, the elementary square inequality gives

$$bT + \frac{aN\kappa_N}{T} \geq 2\sqrt{abN\kappa_N}.$$

Consequently,

$$\begin{aligned} \left(\frac{N}{T} + b\right)(T + a\kappa_N) &\geq N + 2\sqrt{abN\kappa_N} + ab\kappa_N \\ &= \left(\sqrt{N} + \sqrt{ab\kappa_N}\right)^2 = \kappa_N^2. \end{aligned} \quad (7.6)$$

Equality occurs at $T = \sqrt{aN\kappa_N/b}$. In particular, κ_N satisfies the entire collection of these scalar inequalities simultaneously, even if one allows every $T > 0$ and removes the nonnegative exponential error. It therefore also satisfies (5.1) for every allowed $0 < T \leq N$.

Formula (7.5) gives the exact remainder identity

$$\kappa_N - x^2 - \gamma x = \frac{\gamma^2}{2} + \frac{\gamma^3}{4(\sqrt{x^2 + \gamma^2/4} + x)}. \quad (7.7)$$

Thus $\kappa_N = x^2 + \gamma x + O(1)$. Even imposing integrality on a scalar k does not change this leading limitation: for each fixed T , the inequality in k is a monic quadratic inequality with negative constant term, so it holds on the full interval from zero to its positive root. It therefore holds at $\lfloor \kappa_N \rfloor$, simultaneously for all T , including with the exponential error restored. These integers still equal $x^2 + \gamma x + O(1)$. For large N they also lie between 0 and N .

It follows that these scalar inequalities alone cannot imply $k \leq \sqrt{N} + cN^{1/4} + C$ for any fixed $c < \gamma$ and fixed C : the admissible scalar values $\lfloor \kappa_N \rfloor$ eventually violate that conclusion. These are scalar comparison values, **not constructed Sidon sets**. Improving an earlier estimate, retaining more information about the differences, or choosing another kernel lies outside this limitation statement. No general method barrier is claimed.

8 Remarks

8.1 A numerical kernel scan: non-rigorous evidence

An exploratory discretization considered the minimum energy of signed measures of mass one supported on $[0, L]$, for $L = 4, 8, 16$, with approximately 100–200 grid points per unit interval. It used $1/E_{\min} - L$ as an estimate of a boundary constant b and compared $\sqrt{ab}$, where $a = f(0)$. For the normalized weights $h_p(t) = (p+1)(1-t)^p \mathbf{1}_{[0,1]}(t)$, the recorded approximate scores were

p	0	1/2	0.8	1	1.2	1.5	2
$\sqrt{ab}$ (reported)	1.0000	0.9487	0.9435	0.9428	0.9433	0.9449	0.9487.

The linear weight had a sampled boundary estimate 0.6687, compared with the exact $2/3$ proved above. The scan also reported scores near 0.9470 and 0.9446 for normalized cosine and squared-cosine weights. These are floating-point observations, without certified discretization or rounding errors. They do not establish a local minimum in a continuous family, optimality over kernels, or a limit of a general method. None of these observations is used in the theorem or in the scalar limitation statement of Section 7.

8.2 The same numerical constant in a different ruler problem

Gupta and O'Bryant [GO26, Definition 2 and Theorem 3] define an LM ruler by requiring that a positive distance d have at most $d - 1$ representations. For an n -element LM ruler $\{0 = \ell_1 < \dots < \ell_n\}$ they prove

$$\ell_n \geq \sqrt{8/9} (n-1)^{3/2}.$$

Thus the number $\sqrt{8/9} = 2\sqrt{2}/3$ occurs there as well. This is a numerical coincidence between the stated constants: the multiplicity condition, main scale, and conclusion differ. No implication between that LM-ruler theorem and the Sidon cardinality bound of this note is asserted.

8.3 Analytic inputs and finite checks

The proof uses only standard real analysis: Tonelli and Fubini under the nonnegativity and absolute-integrability conditions verified in the text; monotone and dominated convergence; Cauchy–Schwarz in real L^2 ; absolute continuity of indefinite integrals of locally integrable functions and its integrating-factor consequence; completeness of the real numbers for nested compact intervals; and elementary exponential, logarithmic, and quadratic calculations. The factorial majorant proves compact uniform convergence of the density series. No external result about Sidon sets, renewal processes, or extremal constants is an input to the proof.

The accompanying audit script `check_sidon_bound.py` uses exact fractions and integer square roots for rational enclosures and finite sanity checks. A JavaScript translation was executed for the reported finite checks, including maxima on intervals of lengths at most 60; the Python deliverable itself was not executed in the drafting environment. These finite computations are not used to prove the theorem. In particular, numerical grids do not replace the all- N monotonicity argument in Section 6.

Provenance and verification status

The argument was found by a clean-room GPT-6 Astra agent (OpenAI), then expanded and examined in informed same-vendor adversarial reviews. The source proof, the standalone expansion, and the finite-audit code received same-vendor review; this is the completed mathematical review status of the present draft. Cross-vendor referee reports by Claude Opus (Anthropic): two isolated reports dated 2026-10-02, both *PASS* with no repair required: referee A re-derived every display and checked the exact potential of the capacity lemma numerically; referee B reconstructed the main inequality independently from the statement and the constant, and tested every intermediate inequality on about $1.2 \cdot 10^5$ Sidon sets (all maximum Sidon sets for $N \leq 60$, Singer and Bose–Chowla sets for prime powers $q \leq 127$, Mian–Chowla prefixes, random and adversarial sets) without a violation. Both referees note that the onset 120^4 is a convenient sufficient value: the same inequalities already give the stated bound for $N \geq 4\,540\,589$ (numerical evaluation, not certified here). No independent human referee report, Lean formalization, or other kernel verification is claimed. The author takes responsibility for the mathematics. Bibliographic statements and unrefereed numerical claims are distinguished in the introduction; citing a claim does not certify its proof or computational certificate.

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